lived in Carolina, Bryce DeWitt, provided the key.[4] Wheeler travelled frequently, and met collaborators wherever he could. He asks Bryce to meet at Raleigh Durham airport, in North Carolina, where he had a few hours' wait between connecting flights. Bryce arrives and shows him an equation for 'a wave function of space', obtained by using a simple mathematical trick.[fn31] Wheeler is enthused. From this conversation a type of 'equation of orbitals' for general relativity is born; an equation which should determine the probability of one or another curved spaces. For a long time, DeWitt called it Wheeler's equation[fn32] – while Wheeler called it the DeWitt equation. Everyone else calls it the Wheeler-DeWitt equation.

The idea is very good, and becomes a basis for the attempts to construct the full theory of quantum gravity. But the equation itself is riddled with problems – serious ones. In the first place, from a mathematical point of view, the equation is really quite badly defined. If we try to use it to do calculations, we soon obtain results that are infinite, which makes no sense. It must be improved.

But it is also difficult to understand how to interpret this equation, to know what it means. Among its disconcerting aspects is the fact that it no longer contains the time variable. How can it be used to compute the evolution of something which happens in time if it does not include a time variable? Dynamical equations, in physics, always contain the variable *t*, time. What does a physical theory without a temporal variable signify? For years, research will revolve around such questions, trying to revise the equation in different manners, in order to improve its definition and understand what it might mean.

The first steps of the loops

The fog begins to dissipate towards the end of the 1980s. Surprisingly, some solutions of the Wheeler–DeWitt equation appear. During these years I found myself first at the University of Syracuse, in New York State, visiting the Indian physicist Abhay Ashtekar, and then in Connecticut, at Yale University, visiting the American physicist Lee Smolin. I remember a period of intense discussions and burning intellectual fervour. Ashtekar had rewritten the Wheeler–De Witt equation in a simpler form; and Smolin, together with Ted Jacobson of the University of Maryland in Washington, had been the first to find some of these new strange solutions of the equation.

The solutions had a curious peculiarity: they depended on *closed lines* in space. A closed line is a 'loop'. Smolin and Jacobson could write a solution to the Wheeler–DeWitt equation for every loop: for every line closed on itself. What did this mean? The first works of what will later become known as loop